TEMITOPE AKINBOYEWA

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EDUCATION

- Ph.D Student, Geography, The Pennsylvania State University, University Park, PA, USA [Aug. 2023 – present (*expected to graduate in Spring 2027)*]
- M.S. Remote Sensing and GIS, The Federal University of Technology, Akure, Nigeria [Jan. 2021- Jul. 2023]
- B.S. Surveying and Geoinformatics, The Federal University of Technology, Akure, Nigeria [Jan. 2014 – Dec. 2019]

PROFESSIONAL EXPERIENCE

- 2023 – present: Graduate Research and Teaching Assistant, Department of Geography, Pennsylvania State University, PA, USA
- 2026: AI Research Intern, bAI Labs, Annapolis, MD, USA
- test and validate the reliability and field-readiness of AI models and agent-based tools within applied environmental engineering scenarios.
- 2022 – 2023: Graduate Research and Teaching Assistant, Department of Surveying and Geoinformatics, Federal University of Technology, Akure, Ondo State, Nigeria
- 2021 – 2022: Field surveyor and Geospatial Data Analyst, Ademag and Associates (Surveying and Geoinformation), Akure, Nigeria
- 2020 – 2021: GIS/ Remote Sensing Analyst, Kaduna State Urban Planning and Development Authority (KASUPDA), Kaduna, Nigeria

PUBLICATIONS

Peer-reviewed Journal articles

1. Li, Z., Khosravi Kazazi, A., Liu, R., **Akinboyewa, T.**, Ning, H., Huang, X., Ye, X., Arundel, S. T., Li, W., Yang, C., Wang, S., & Simonis, I. (2026). Geospatial Agentic Services: A framework for interoperable geospatial intelligence. Annals of GIS, 1–21. https://doi.org/10.1080/19475683.2026.2738374

2. **Akinboyewa, T.**, Ning, H., Li, Z., Lessani, M. N., Li, X., & Qiao, S. (2026). Exploring the role of place visitation big data on small area health measure estimation. Computational Urban Science, 6(1), 52. https://doi.org/10.1007/s43762-026-00287-0
3. **Akinboyewa, T.**, Li, Z., Ning, H., Lessani, M. N., Holmes, L. M., & Qiao, S. (2025). Exploring the association between smartphone-based place visitation data and neighborhood-level coronary heart disease in the United States. *PLOS One*, 20(8), e0329455. https://doi.org/10.1371/journal.pone.0329455
4. **Akinboyewa, T.**, Li, Z., Ning, H., & Lessani, M. N. (2025). GIS Copilot: Towards an autonomous GIS agent for spatial analysis. International Journal of Digital Earth, 18(1). https://doi.org/10.1080/17538947.2025.2497489
5. Ning, H., Li, Z., **Akinboyewa, T**., & Lessani, M. N. (2025). An Autonomous GIS Agent Framework for Geospatial Data Retrieval. *International Journal of Digital Earth*. https://doi.org/10.1080/17538947.2025.2458688
6. **Akinboyewa, T**., Ning, H., Lessani, M. N., & Li, Z. (2024). Automated floodwater depth estimation using large multimodal model for rapid flood mapping. *Computational Urban Science*, *4*(1), 12. https://doi.org/10.1007/s43762-024-00123-3
7. Nzelibe, I. U., **Akinboyewa, T. E**., Nzelibe, T. N., & Inekwe, G. G. (2022). Geospatial Assessment of The Potentials of Rooftop Rainwater Harvest at The Federal University of Technology, Akure, Nigeria. *FUOYE Journal of Engineering and Technology*, *7*(2), 249-256. https://doi.org/10.46792/fuoyejet.v7i2.839

Book chapter

1. **Akinboyewa, T**., & Li, Z. (2026). GeoAI and AI Copilot. In D. Richardson, N. Castree, M. F. Goodchild, A. Kobayashi, W. Liu, & R. A. Marston (Eds.), *International Encyclopedia of Geography* (1st ed., pp. 1–8). Wiley. https://doi.org/10.1002/9781118786352.wbieg2254

Journal articles under review

1. **Akinboyewa, T.**, & Li, Z. (2026). Toward Self-growing Data Retrieval Capabilities in Autonomous GIS: A Framework for Geospatial Data Source Onboarding. SSRN. https://doi.org/10.2139/ssrn.7527681
2. Lessani, M. N., Li, Z., **Akinboyewa, T.**, & Jing, F. (2026). Place-based Representation Bias in Human Mobility Data: Assessing POI Coverage, Category Bias, and Visitor Representation Bias. SSRN. https://doi.org/10.2139/ssrn.7314098
3. **Akinboyewa, T**., & Li, Z. (2026). GIS Co-Scientist: A Human-AI Collaborative Framework for Autonomous Geospatial Scientific Inquiry. (Under review). Preprint: https://dx.doi.org/10.2139/ssrn.6753098

4. **Akinboyewa, T.**, Li, Z., Qiao, S., Holmes, L. M., Yu, M., & Kaczynski, A. T. Examining differences in COVID-19 impact on dietary behavior across the United States using cellphone-based place visitation data. (Under review)

5. Lim, S., Khosravi Kazazi, A., **Akinboyewa, T.**, & Li, Z. Unequal Adaptation: Threshold Effects of Micromobility Substitution under Extreme Heat in New York City. (Under review)

KEY INNOVATIONS & OPEN-SOURCE AI TOOLS

- **GIS Co-Scientist | Lead developer (2026)**
  - Developed a human-AI collaborative framework that autonomously formulates hypotheses, designs analytical workflows, and executes spatial analyses to accelerate geographic scientific discovery.
- QGIS Plugin: SpatialAnalysisAgent |GeodataRetrieverAgent
  - Designed and developed the first AI-driven Spatial Analysis and Geodata Retrieval agent integrated into QGIS, a leading open-source GIS platform used globally by professionals and researchers.

PRESENTATIONS

1. Towards Autonomous Geospatial Scientific Inquiry: An AI Co-Scientist Framework for Geographic Knowledge Discovery, *AI and Social Research: Empathic AI, Metascience, and Methodology*, June 3, 2026, Pennsylvania State University, University Park, PA, USA (**Akinboyewa, Temitope**)
2. Towards Autonomous Geospatial Scientific Inquiry: An AI Co-Scientist Framework for Geographic Knowledge Discovery, *IndustryXchange 2026: Engineering Innovation with AI*, May 19, Pennsylvania State University, University Park, PA, USA (**Akinboyewa, Temitope**) – 1st Place Poster Award
3. Multimodal Geospatial Data Fusion for Flood Depth Prediction Across the Contiguous United States, *NSF I-GUIDE Summer School*, team project presentation), August 4-8, Boulder, CO, USA (Ghaneei, P., Berg, A., Balogun, R. O., Liu, J., **Akinboyewa, T.**, Chen, X., Zhou, W.) https://platform.i-guide.io/oers/4aaf4262-4193-4b66-a88a-f6d8ae782dae
4. Examining Differences in COVID-19 Impact on Dietary Behavior Across the United States Using Cellphone-Based Place Visitation Data, *2026 PA GIS Conference* (**Akinboyewa, Temitope**)
5. Autonomous Geospatial Modeling: Towards Geographic Discovery, *2026 AAG Annual meeting*, March 17-21, San Francisco, CA, USA (**Akinboyewa, Temitope**)

6. GIS Copilot: Towards an Autonomous GIS Agent for Spatial Analysis, *2025 AAG Annual meeting*, March 24-28, Detroit, MI, USA (**Akinboyewa, T.**, Li, Z., Ning, H., Lessani, M. N.)
7. Smartphone-based Place Visitation Data Better Explain Neighborhood-level Coronary Heart Disease in USA, *2024 AAG Annual meeting* (**Akinboyewa, T.**, Ning, H., Lessani, M. N., Li, Z.)
8. Mapping Minds Shaping the World, 2024 GIS day Webinar, Organized by the Nigerian Institution of Surveying and Geoinformatics Students (**Akinboyewa, T**)
9. GIS Copilot: Towards an Autonomous GIS Agent for Spatial Analysis, *2024 Middle States Division AAG*, West Chester, PA, USA **(Akinboyewa, T.**, Li, Z., Ning, H., Lessani, M. N.)
10. Automated floodwater depth estimation using large multimodal model for rapid flood mapping, *AAG 2024 Penn State Node* (**Akinboyewa, T**)

ACADEMIC SERVICES AND OUTREACH

Conferences Sessions Organized/Chaired

1. AAG 2026 GeoAI and Deep Learning Symposium: The Convergence of Generative AI and GIScience: Domain Applications and Use Cases, AAG Annual Meeting, March 17-21, 2026, San Francisco, CA. (**Chair)**
2. AAG 2026 GeoAI and Deep Learning Symposium: The Convergence of Generative AI and GIScience: Benchmarking, Fine-Tuning, and Evaluation, AAG Annual Meeting, March 17-21, 2026, San Francisco, CA. (**Chair**)
3. AAG 2026 Symposium on Spatial AI and Data Science: Frontiers and Applications: Understanding Human–Environment Interaction Using Geospatial Big Data and GeoAI, AAG Annual Meeting, March 17-21, 2026, San Francisco, CA. (**Chair**)
4. Symposium on Spatial AI and Data Science: Frontiers and Applications: Integrative Approaches to Understanding Human Mobility and Health Outcomes, AAG Annual Meeting, March 17-21, 2026, San Francisco, CA. (**Chair**)
5. AAG 2025 Symposium on Spatial AI & Data Science for Sustainability: Integrative Approaches to Understanding Human Mobility and Health Outcomes, AAG Annual Meeting, March 24-28, 2025, Detroit (**Chair**, co-organized with Ning, H., Lessani, M. N., Li, Z.).
6. Integrative Approaches to Understanding Human Mobility and Health Outcomes, AAG Annual Meeting, April 16-20, 2024, Honolulu (**Chair**, co-organized with Ning, H., Lessani, M. N., Li, Z., Yu, M., held virtually)

7. Geospatial Big Data for Analyzing and Understanding Human Mobility Patterns, April 16-20, 2024, Honolulu (**Chair**, co-organized with Lessani, M. N., Li, Z., Ning, H., Yu, M., held virtually)

Manuscript Reviews for Peer-Reviewed Journals

- International Journal of Digital Earth
- PLOS ONE
- Cartography and Geographic Information Science
- International Journal of Geographical Information Science
- International Journal of Applied Earth and Geoinformation
- Transactions in GIS
- Geo-spatial Information Science
- Structural Control and Health Monitoring

Manuscript Reviews for Conferences

- Deep learning Indaba
- NSF NHERI GSC Mini-Conference

HONORS AND AWARDS

- 2026: J. Jeffrey and Ann Marie Fox Graduate School Fox Degree Completion Award (Fox Scholar), 2025–26 academic year, Pennsylvania State University, PA, USA. – $10,000
- 2026: 1st Place Poster Award, IndustryXchange 2026: Engineering Innovation with AI, Pennsylvania State University, PA, USA."
- 2026: Outstanding Graduate Research Assistant Award, Department of Geography, Pennsylvania State University, PA, USA. - $700
- 2025: I-GUIDE Summer School 2025 Travel Award (NSF-Funded) -$2,685.68
- 2025: Top 5 Finalist, CISG Robert Raskin Paper Competition, USA
- 2024: Outstanding Graduate Teaching Assistant Award, Department of Geography, Pennsylvania State University, PA, USA - $800
- 2023: Outstanding Employee, Ademag and Associates (Surveying and Geoinformation), Akure, Nigeria
- 2017: Winner of the annual inter-tertiary Oratory contest, National Union of Ileoluji Student, Nigeria

TECHNICAL EXPERIENCE

- **AI / Machine Learning**: PyTorch, TensorFlow, Keras, Large Multimodal Models (LMMs), Large Language Models (LLMs), Vector Databases, Multi-Agent Orchestration, Human-in-the-Loop (HITL) Architecture, AI Governance Frameworks, Prompt engineering, Agentic AI
- **Geospatial Data Science**: Python (GeoPandas, NumPy, Pandas, Scikit-learn), QGIS, ArcGIS Pro, AutoCAD, ENVI, ERDAS Imagine, Spatial Analytics, Statistical Modeling, Satellite imagery processing.
- **Field Engineering & Surveying**: Differential GPS (DGPS), Total Station, Digital/Optical Level, high accuracy geodata collection, and asset mapping.
- **Platforms & Environments**: Ubuntu/Linux, Windows, VS Code, PyCharm, Conda Environment Management, Cloud Computing Frameworks

PROFESSIONAL MEMBERSHIPS

- 2026-present: Keystone GIS
- 2025-present: Cartographic and Geographic Information Society
- 2025-present: Cyberinfrastructure Specialty Group, AAG
- 2023-present: Association of the American Geographers (AAG)

ORGANIZATION SERVICE

- 2024-2025, Grad Student Representative, Department of Geography, Pennsylvania State University.

MEDIA MENTIONS

1. 2025, An Industry First? – Pulling together QGIS and LLM for GIS tasks, https://www.linkedin.com/posts/jamie-whittall_an-industry-first-pulling-together-qgis-activity-7284928587470700545-f02M
2. 2025, GIS Copilot: An autonomous GIS agent for spatial analysis in QGIS, https://mappinggis.com/2024/11/gis-copilot-un-agente-gis-autonomo-para-el-analisis-espacial-en-qgis/